When Does Self-Supervision Beat ImageNet?

Domain Distance Governs Label-Efficient Pretraining in Medical Imaging

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Abstract

Labelled medical images are scarce because annotation requires clinician time, yet self-supervised pretraining promises to convert unlabelled scans into useful representations. It is usually unclear *a priori* whether a team should invest in in-domain self-supervision or simply reuse ImageNet weights. We study this choice directly by holding the architecture, optimiser, augmentation budget and evaluation protocol fixed and varying only the initialisation of a ResNet-18 across four arms — random, ImageNet transfer, in-domain SimCLR, and sequential ImageNet→SimCLR — over five label budgets (1–100%), three seeds and two MedMNIST v2 tasks that differ in how far they sit from natural images: grayscale paediatric chest radiographs (PneumoniaMNIST) and RGB dermatoscopy (DermaMNIST). Across 210 supervised runs we find that the ranking of the two pretraining strategies *inverts* with domain distance. On chest X-ray at a 1% budget (47 labelled images) in-domain SimCLR reaches *0.971* AUROC against *0.917* for ImageNet transfer and *0.929* for random initialisation, a paired gain of *+0.042* over scratch (*p=0.011*); ImageNet transfer is not significantly better than scratch on this task at any budget. On dermatoscopy the opposite holds: ImageNet transfer leads at every budget and in-domain SimCLR never separates from scratch. With the full label set every pretraining advantage disappears (*p>0.14*), confirming that these methods buy label efficiency rather than peak accuracy. An augmentation ablation explains part of the asymmetry: removing colour jitter from the SimCLR view generator *improves* dermatoscopy (*0.798→0.823*) while degrading radiography (*0.982→0.959*), because colour carries diagnostic signal in pigmented lesions but acts only as an intensity augmentation on grayscale film. In-domain SimCLR is also the best-calibrated and the most corruption-robust arm, losing only *0.036* AUROC under severe additive noise where ImageNet transfer loses *0.275*. We conclude that the useful question is not “does self-supervision help?” but “how far is this modality from the pretraining corpus, and does the augmentation family preserve the diagnostic signal?”

1 Problem Formulation

The clinical problem. Supervised deep learning in medical imaging is bottlenecked by labels, not by images. A hospital PACS accumulates studies continuously, but turning them into a training set requires expert annotation that competes directly with clinical service time; for dermatoscopy, ground truth often additionally requires histopathological confirmation [1]. The consequence is the regime this paper studies: thousands of unlabelled in-domain images alongside only tens to hundreds of labelled ones. Lecture 1 framed this as one of the field's defining constraints, and Lecture 2's case studies (Gulshan 2016, Esteva 2017, Rajpurkar 2017) are precisely the exceptions that required extraordinary labelling effort [2,3].

Two competing answers. Practitioners have two standard ways to exploit structure learned elsewhere. The first, covered in Lecture 2–3 under transfer learning, initialises from ImageNet [4,5] and fine-tunes. The second, covered in Lecture 8 on self-supervised and contrastive learning, pretrains on the unlabelled in-domain images themselves with an objective such as SimCLR [6]. Both are routinely reported to help, and both are routinely reported with a single dataset, which makes the practical question — *which one should I spend my compute on?* — unanswerable from the literature.

Research question and hypotheses. We ask whether the better pretraining strategy under label scarcity is determined by the distance between the target modality and the natural-image corpus, and we test four hypotheses: **H1** (scarcity) pretraining of either kind helps most at small label budgets and its advantage vanishes as labels grow; **H2** (domain distance) in-domain SimCLR is favoured on a modality far from natural images (grayscale radiography) whereas ImageNet transfer is favoured on a modality close to them (RGB dermatoscopy); **H3** (augmentation) the optimal SimCLR augmentation family is modality-specific, because an augmentation that destroys a diagnostic cue cannot yield an invariance worth having; **H4** (composition) sequential ImageNet→SimCLR pretraining combines both sources of structure and improves on either alone in the scarce regime. Our contribution is not a new architecture but a controlled answer to a decision that every applied team faces, together with the mechanism behind it.

Use of Software 3.0 in problem formulation. An AI copilot (Claude) was used to survey candidate framings and datasets. Its first proposals were the generic "benchmark SSL versus transfer learning on a medical dataset", which we rejected as a re-run of published results with no hypothesis. The framing we kept — two datasets chosen to *differ in domain distance*, so that the comparison can invert — came from pushing back on that suggestion and asking what result would be informative either way. The copilot also proposed MIMIC-CXR, which we rejected on feasibility grounds (PhysioNet credentialing takes weeks), and the augmentation-ablation design of H3, which we kept and sharpened by predicting the *sign* of the colour-jitter effect per modality before running it. We treat this as the honest record of AI assistance: useful for breadth and for naming standard practice, unreliable as a source of a sharp question.

2 Related Work

Transfer learning in medical imaging. ImageNet initialisation is the default in medical computer vision [8] and underpins the landmark results discussed in Lecture 2 [2,3]. Raghu et al. [7] questioned the mechanism: for medical targets, much of the benefit of ImageNet weights comes from better-conditioned scaling of the early layers rather than from reusable high-level features, and small purpose-built networks often match transferred ones given enough labels. Our 100% column reproduces that observation and extends it by showing the regime where it fails — at 1% labels the initialisation matters enormously.

Self-supervised representation learning. SimCLR [6] and MoCo [10] learn by making two augmented views of the same image agree, with the augmentation family defining the invariances learned. Lecture 8 (Contrastive Learning: SimCLR; Weak Supervision) presents exactly this framework, including the observation that augmentation choice, not architecture, drives representation quality — the claim H3 tests in a medical setting. Azizi et al. [9] showed that large self-supervised models improve dermatology and chest X-ray classification, particularly when in-domain pretraining follows ImageNet pretraining; our H4 arm is a small-scale version of that recipe, and our robustness analysis shows a failure mode of it that accuracy alone hides.

Multimodal and text-supervised pretraining. ConVIRT [11] and GLoRIA [12], both covered in Lecture 8–9 alongside CLIP [13], exploit paired radiology reports as free supervision and are the strongest label-efficient methods for chest X-ray. They require paired text, which MedMNIST does not provide; our image-only setting is the regime a team faces when reports are inaccessible, and our X-ray result (in-domain contrastive pretraining being decisively better than ImageNet) is consistent with their premise that in-domain structure is what matters for radiography.

Benchmarks, calibration and robustness. MedMNIST v2 [17] standardises lightweight 2D/3D medical classification and provides the ResNet-18 reference numbers we sanity-check against; its source datasets are HAM10000 [1] and the paediatric chest X-ray collection of Kermany et al. [14]. Because accuracy alone is inadequate for clinical deployment, we additionally report expected calibration error [18] and corruption robustness in the style of Hendrycks et al. [15], both themes of

Lecture 11 (Interpretability, Fairness, Uncertainty, Robustness), and we interpret models with Grad-CAM [19], the attribution method introduced in that same lecture. Our optimiser follows Loshchilov et al. [16] and our GPU augmentation uses Kornia [20].

Where this work sits. Prior work establishes that both pretraining strategies help, usually one dataset at a time. We hold everything but the initialisation fixed, evaluate two modalities chosen to differ in domain distance, and report label efficiency, calibration, robustness and augmentation sensitivity together. The result — that the ranking inverts, and that the augmentation that helps one modality hurts the other — is a statement about *when* each method applies rather than which is better.

3 Data

We use two MedMNIST v2 [17] tasks at *64×64* resolution with the official train/validation/test splits, which are patient-disjoint in the source datasets.

PneumoniaMNIST derives from paediatric chest radiographs [14]: 4,708 training, 524 validation and 624 test images, grayscale, binary (normal versus pneumonia), with 74% of training images positive. **DermaMNIST** derives from HAM10000 [1]: 7,007 training, 1,003 validation and 2,005 test images, RGB, seven lesion classes with severe imbalance — melanocytic nevi account for 4,693 training images while dermatofibroma has 80. Figure 1 shows examples and the full class distribution.

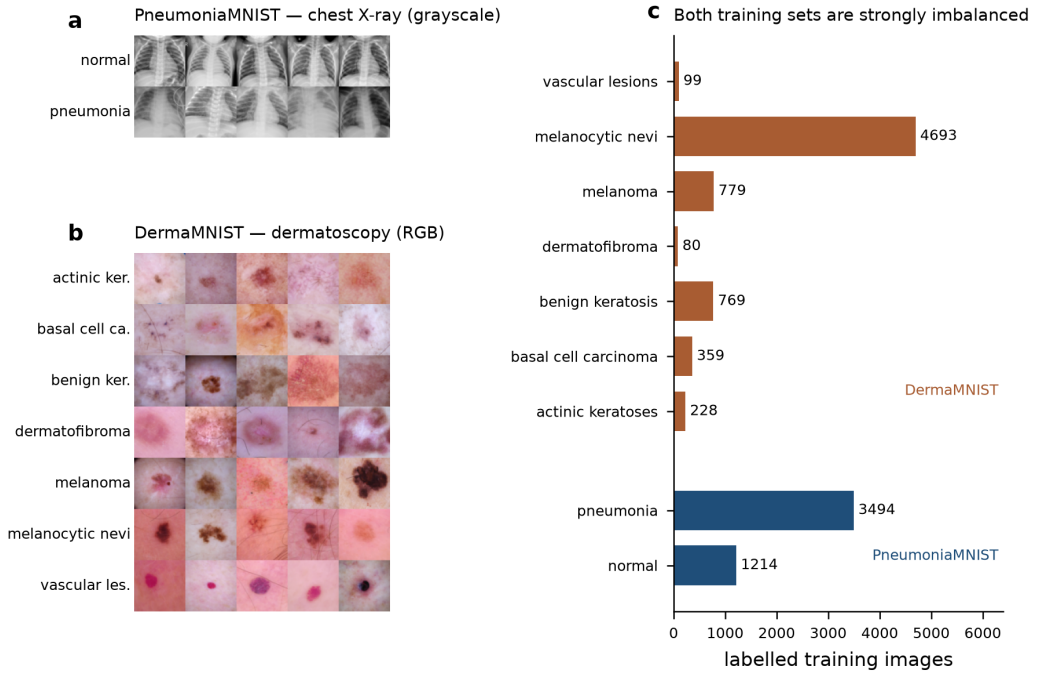


Figure 1: The two tasks. (a–b) Five random training examples per class; the modalities differ in channel statistics, in what constitutes the diagnostic cue (opacity and lung texture versus lesion pigmentation and border), and in how closely they resemble natural images. (c) Training-set class counts; both tasks are strongly imbalanced, which is why every model is trained with a class-weighted loss and evaluated with balanced accuracy and AUPRC in addition to AUROC.

Label budgets. The label-scarcity axis is created by class-stratified subsampling of the training split at fractions *1,5,10,25,100%*. Subsets are *nested* within a seed (the 1% subset is contained in the 5% subset, verified programmatically), so differences along the curve reflect the budget rather than which images happened to be drawn, and every class retains at least five examples so that no class disappears. The 1% budget corresponds to 47 labelled chest radiographs and 83 dermatoscopic images — the latter spread

over seven classes, i.e. 5–56 images per class. Validation and test splits are always complete; only the training labels are restricted, which mirrors the real situation of having abundant unlabelled data.

Unlabelled data for pretraining. Self-supervised pretraining uses the images of the full training split with labels discarded (4,708 and 7,007 images). This is deliberately a *small* unlabelled corpus by self-supervised standards and bounds how much in-domain SimCLR can be expected to achieve.

Preprocessing. Images are kept at 64×64, scaled to $[0,1]$, and normalised with ImageNet channel statistics in every arm so that normalisation is not a confound. Grayscale images are replicated to three channels so that one unmodified architecture serves all arms.

4 Approach

Design principle. The experiment is designed so that the only difference between arms is the *initialisation* of the backbone. Architecture, input resolution, normalisation, supervised augmentation, loss, optimiser, schedule, early stopping and model selection are identical across arms, which is what licenses reading a difference in test metric as an effect of pretraining.

Backbone. An unmodified torchvision ResNet-18 [5], including its 7×7/stride-2 stem, fed with 64×64 inputs (final feature map 2×2, 512-dimensional pooled embedding). Keeping the ImageNet stem rather than substituting a CIFAR-style 3×3 stem costs some spatial resolution but is the only way to reuse ImageNet weights without surgery, and therefore the only way to keep the architecture common to all four arms.

Arms. (i) *From scratch*: random initialisation. (ii) *ImageNet transfer*: `IMAGENET1K_V1` weights. (iii) *In-domain SimCLR*: SimCLR pretraining on the unlabelled in-domain images from random initialisation. (iv) *ImageNet + SimCLR*: the same SimCLR pretraining started from ImageNet weights.

SimCLR pretraining. Two views per image are drawn from the standard SimCLR family — random resized crop (scale $0.2–1.0$), horizontal flip, colour jitter $(0.8,0.8,0.8,0.2)$ with probability 0.8, random grayscale ($p=0.2$) and Gaussian blur (5×5, $\sigma\in[0.1,2]$, $p=0.5$). A two-layer MLP head ($512\to512\to128$) projects the pooled features, and the NT-Xent loss [6] with temperature $\tau=0.5$ treats the other view as the positive and the remaining $2N-2$ elements of the batch as negatives:

$$L_i = -\log \frac{\exp(\mathrm{sim}(z_i, z_j(i))/\tau)}{\sum_{k \neq i} \exp(\mathrm{sim}(z_i, z_k)/\tau)}, \quad \mathrm{sim}(u,v) = u^T v / \|u\| \, \|v\|.$$

We pretrain for 400 epochs with AdamW [16] (learning rate 10^{-3}, weight decay 10^{-6}), batch size 256 and a cosine schedule. The loss is computed in float32 with autocast disabled, because under mixed precision the self-similarity mask overflows the float16 range — a concrete implementation detail that silently produces NaNs if missed.

Supervised training. Each labelled subset is trained for at most 40 epochs with AdamW (learning rate 3×10^{-4} for pretrained initialisations, 10^{-3} for random initialisation and for linear probes; weight decay 10^{-4}), batch size $\min(128, \max(8, N/4))$, cosine schedule, and augmentation limited to random resized crop (scale $0.7–1.0$) and horizontal flip. The loss is cross-entropy with inverse-frequency class weights and label smoothing 0.05. We select the epoch with the best *validation* AUROC (patience 8) and evaluate the test split exactly once with those weights, so no test information enters model selection. Linear probes freeze the backbone (kept in eval mode so its batch-norm statistics are not updated) and train only the final linear layer.

Metrics. We report AUROC and AUPRC (macro one-vs-rest for the seven-class task), balanced accuracy, macro-averaged sensitivity and specificity, and expected calibration error with 15 equal-width confidence bins [18]. Balanced accuracy and AUPRC are emphasised because accuracy is uninformative when 67% of dermatoscopy images belong to one class. Every configuration is run with seeds $0,1,2$ and we report the mean with a 95% t-interval; comparisons against the scratch arm use a paired t-test over seeds, pairing on the seed that also determines the label subset.

Implementation and Software 3.0. PyTorch 2.11 (CUDA 12.8) on one RTX 5090; both datasets are held resident in GPU memory as tensors and augmented in batch with Kornia [20], with per-sample augmentation parameters. This was a necessary engineering decision rather than an optimisation: the execution sandbox denies named-pipe creation, so multi-process `DataLoader` workers are unavailable and in-process CPU augmentation capped throughput at 8–12 s per pretraining epoch; moving augmentation onto the device reduced this to 0.8–1.2 s and made 12 pretraining runs and 210 supervised runs feasible in about two hours. An AI copilot wrote much of the scaffolding (data module, metric functions, sweep driver) and diagnosed two failures that cost real time — the float16 overflow in NT-Xent and an OpenMP runtime clash between a conda MKL NumPy and the venv PyTorch. It was least reliable where it mattered most: its initial experiment driver evaluated on the test split for model selection, which we replaced with validation-based selection. We report this because it is the pattern we observed throughout — AI assistance compressed implementation time substantially while leaving experimental validity entirely to the authors.

5 Experiments

All numbers are test-set means over three seeds with 95% *t*-intervals; * marks $p<0.05$ in a paired *t*-test against the scratch arm.

5.1 Label efficiency and the inversion (H1, H2)

Table 1 and Figure 2 give the main result. Both classical baselines are beaten by every neural arm on chest X-ray, but on dermatoscopy a pixel random forest (*0.807* AUROC at 100% labels) is a stronger baseline than one might expect and exceeds every neural arm at the 1% budget — a useful reminder that deep models are not automatically preferable when labels are very scarce.

Table 1: Test AUROC by initialisation and label budget (mean ± 95% CI over three seeds). The pixel random forest is the non-neural reference.

Dataset	Initialisation	Labelled fraction of the training set				
		1%	5%	10%	25%	100%
PneumoniaMNIST	pixel random forest	0.909 ± 0.022	0.931 ± 0.030	0.932 ± 0.022	0.940 ± 0.012	0.944 ± 0.002
	from scratch	0.929 ± 0.026	0.957 ± 0.027	0.936 ± 0.026	0.971 ± 0.005	0.979 ± 0.016
	ImageNet transfer	0.917 ± 0.020	0.953 ± 0.085	0.954 ± 0.036	0.984 ± 0.013	0.985 ± 0.018
	in-domain SimCLR	0.971 ± 0.028	0.983 ± 0.011	0.982 ± 0.019	0.986 ± 0.016	0.985 ± 0.005
	ImageNet + SimCLR	0.980 ± 0.002	0.990 ± 0.002	0.987 ± 0.006	0.994 ± 0.002	0.976 ± 0.030
DermaMNIST	pixel random forest	0.683 ± 0.046	0.710 ± 0.011	0.744 ± 0.015	0.763 ± 0.010	0.807 ± 0.004
	from scratch	0.754 ± 0.072	0.744 ± 0.130	0.795 ± 0.015	0.843 ± 0.061	0.880 ± 0.051
	ImageNet transfer	0.753 ± 0.039	0.780 ± 0.027	0.823 ± 0.057	0.845 ± 0.027	0.914 ± 0.049
	in-domain SimCLR	0.738 ± 0.063	0.765 ± 0.039	0.798 ± 0.062	0.840 ± 0.025	0.887 ± 0.030
	ImageNet + SimCLR	0.743 ± 0.042	0.773 ± 0.038	0.818 ± 0.025	0.855 ± 0.043	0.891 ± 0.038

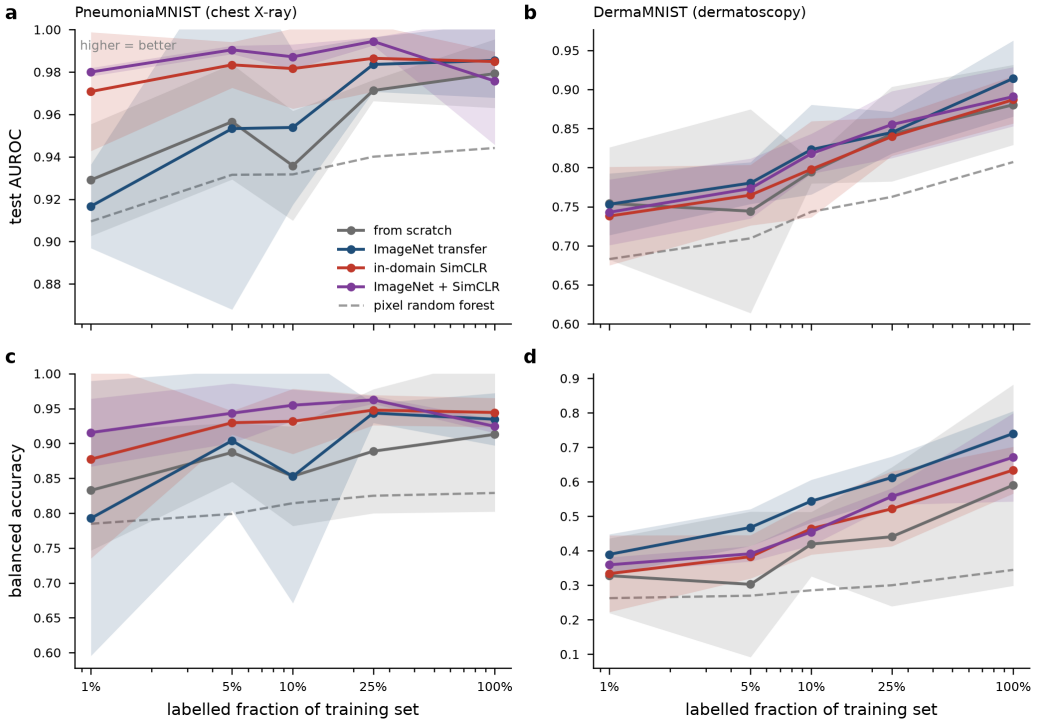


Figure 2: Label-efficiency curves. (a–b) AUROC and (c–d) balanced accuracy against the labelled fraction. On chest X-ray the two SimCLR arms dominate from 1% labels onward and ImageNet transfer tracks the scratch arm; on dermatoscopy the ordering reverses, with ImageNet transfer best at every budget in balanced accuracy. Shaded bands are 95% *t*-intervals over three seeds; the dashed line is the pixel random forest.

The ranking inverts with domain distance (H2, supported). On PneumoniaMNIST at 1% labels (47 images) in-domain SimCLR reaches *0.971* AUROC and sequential ImageNet+SimCLR *0.980*, against *0.917* for ImageNet transfer and *0.929* for scratch. The paired gains over scratch in Table 2 are *+0.042* (*p=0.011*) and *+0.051* (*p=0.012*); ImageNet transfer is *not* significantly better than scratch at any budget on this task, and is numerically worse at 1%. On DermaMNIST the pattern reverses: ImageNet transfer leads at every budget, most visibly in balanced accuracy (*0.544* versus *0.464* for in-domain SimCLR at 10% labels, and *0.740* versus *0.634* at 100%), while in-domain SimCLR never separates from scratch (*+0.003*, *p=0.84* at 10%). The effect we hypothesised is therefore present with the predicted sign in both directions: in-domain self-supervision wins where the modality is far from natural images, ImageNet transfer wins where it is close.

Pretraining buys label efficiency, not peak accuracy (H1, supported). At the full label budget no arm is significantly better than scratch on either task (all *p>0.14*; Table 2). The benefit is concentrated at small budgets, and on chest X-ray it is large enough that the scratch arm never reaches the SimCLR arms' 10% performance at *any* budget we tested, including 100% labels. This is the Raghu et al. [7] observation — initialisation stops mattering once labels are plentiful — with the complementary half made explicit.

Table 2: Paired difference in test AUROC against the from-scratch arm (same seeds, same label subsets). *: *p<0.05*, paired *t*-test, *n=3* seeds.

Dataset	Initialisation	1% labels	10% labels	100% labels
PneumoniaMNIST	ImageNet transfer	-0.012	+0.018	+0.006
	in-domain SimCLR	+0.042*	+0.046*	+0.006
	ImageNet + SimCLR	+0.051*	+0.051*	-0.004

DermaMNIST	ImageNet transfer	-0.001	+0.028	+0.034
	in-domain SimCLR	-0.016	+0.003	+0.007
	ImageNet + SimCLR	-0.011	+0.023	+0.010

Sequential pretraining helps on X-ray only (H4, partially supported). ImageNet+SimCLR is the best arm on PneumoniaMNIST at every budget below 100%, slightly ahead of in-domain SimCLR, so the two sources of structure do compose where self-supervision works. On DermaMNIST it lands between the two single-source arms and does not beat ImageNet transfer. Composition therefore inherits, rather than repairs, the weakness of its in-domain component; and Section 5.4 shows it also inherits a brittleness that neither parent exhibits.

5.2 Linear probes: what the representation already contains

Table 3 separates representation quality from adaptation. On chest X-ray a *frozen* in-domain SimCLR backbone with a single trained linear layer reaches *0.989* AUROC — better than fine-tuning the same weights (*0.982*) and far better than a frozen ImageNet backbone (*0.921*). The pneumonia decision is thus close to linearly decodable from representations learned without a single label, which is the strongest evidence here that in-domain self-supervision has captured task-relevant structure rather than merely a good initialisation. On dermatoscopy no frozen representation is adequate (probes lose to fine-tuning for every arm), consistent with the seven-class fine-grained task requiring feature adaptation.

Table 3: Frozen linear probe versus full fine-tuning at the 10% label budget (test AUROC). A positive final column means the frozen representation is already better than what fine-tuning reaches.

Dataset	Initialisation	Fine-tune	Linear probe	Probe - fine-tune
PneumoniaMNIST	ImageNet transfer	0.954	0.921	-0.033
	in-domain SimCLR	0.982	0.989	+0.007
	ImageNet + SimCLR	0.987	0.991	+0.004
DermaMNIST	ImageNet transfer	0.823	0.815	-0.008
	in-domain SimCLR	0.798	0.776	-0.022
	ImageNet + SimCLR	0.818	0.816	-0.002

5.3 Which augmentation matters, and why it is modality-specific (H3)

Table 4 and Figure 3c ablate one component at a time from the SimCLR view generator, keeping the pretraining budget fixed and fine-tuning at 10% labels.

Table 4: SimCLR augmentation ablation: test AUROC at the 10% label budget after pretraining with one component removed (mean ± 95% CI, three seeds).

Pretraining augmentation	PneumoniaMNIST	DermaMNIST
full SimCLR augmentation	0.982 ± 0.019	0.798 ± 0.062
no colour jitter	0.959 ± 0.034	0.823 ± 0.022
no random resized crop	0.921 ± 0.024	0.725 ± 0.065
no Gaussian blur	0.981 ± 0.027	0.790 ± 0.014
no horizontal flip	0.984 ± 0.012	0.800 ± 0.032

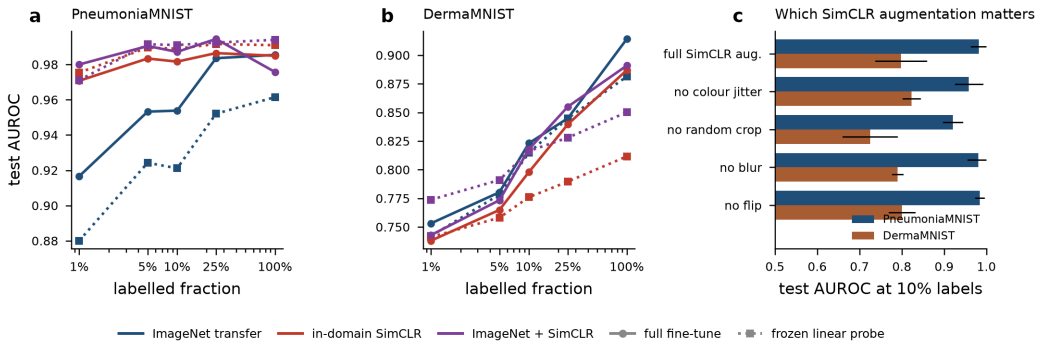


Figure 3: (a–b) Frozen linear probe (dotted) versus full fine-tuning (solid). The in-domain SimCLR probe exceeds its own fine-tuned counterpart on chest X-ray, and the ImageNet probe falls far behind. (c) Augmentation ablation at 10% labels: random resized crop is essential on both modalities, whereas removing colour jitter helps dermatoscopy and hurts radiography.

Random crop is indispensable; colour jitter is not, and its sign flips. Removing random resized crop is the single most damaging change on both tasks (*0.982→0.921* on X-ray, *0.798→0.725* on dermatoscopy): without spatial cropping the contrastive task is too easy and the representation collapses toward low-level cues. Colour jitter behaves oppositely across modalities. Removing it *improves* DermaMNIST from *0.798* to *0.823* — the best dermatoscopy SimCLR result we obtained — but *degrades* PneumoniaMNIST from *0.982* to *0.959*. The interpretation is mechanical and clinically sensible: in pigmented-lesion analysis colour and its variegation *are* diagnostic criteria, so forcing the encoder to be invariant to hue and saturation discards the signal the downstream task needs; on single-channel radiographs the same transform reduces to brightness and contrast perturbation, which mimics real exposure variation and is useful. H3 is supported, and in the strong form: the best augmentation family is not a property of SimCLR but of the modality.

5.4 Calibration, rare classes, robustness and attribution

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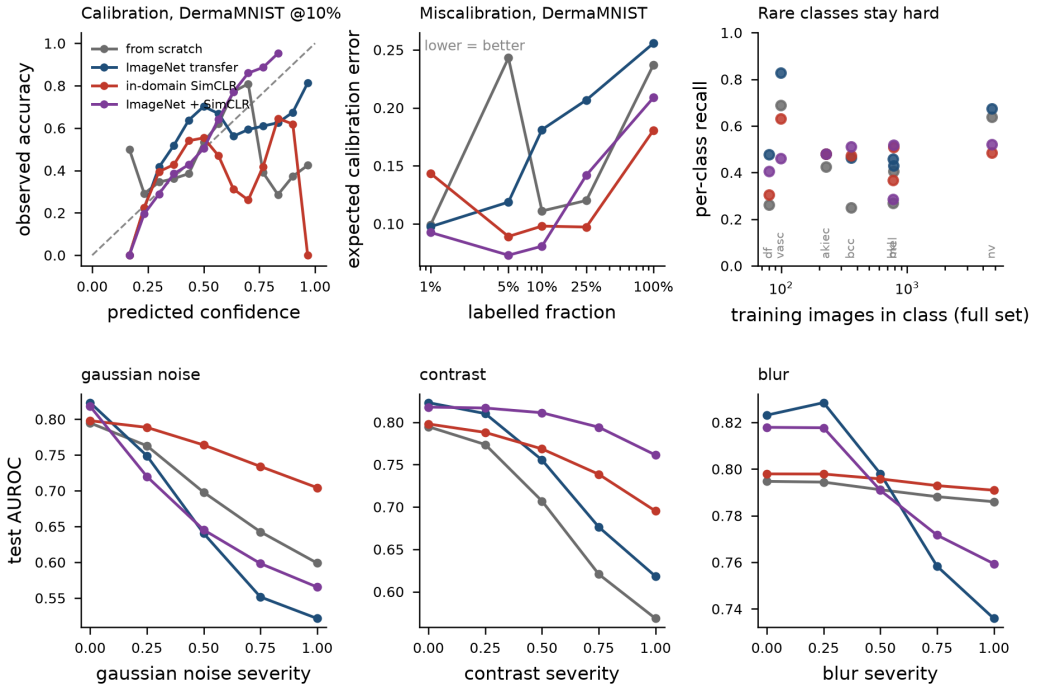


Figure 4: DermaMNIST analyses. (a) Reliability diagram at 10% labels. (b) Expected calibration error against label budget. (c) Per-class recall at 10% labels against class frequency in the full training set; the four rarest classes remain poorly detected by every arm. (d–f) Test-time corruption robustness at the 10% budget, mean over three seeds.

Calibration. At the 10% dermatoscopy budget, expected calibration error is *0.081* for ImageNet+SimCLR and *0.098* for in-domain SimCLR, against *0.181* for ImageNet transfer and *0.111* for scratch. The arm with the best discrimination on this task (ImageNet transfer) is thus the worst calibrated — its confidences are not usable as probabilities without recalibration, which matters more than a 0.02 AUROC difference for any triage application [18].

Rare classes are not rescued by pretraining. Figure 4c plots per-class recall against class frequency at the 10% budget. Recall tracks frequency for every arm, and the rare classes — dermatofibroma (80 training images in the full set) and vascular lesions (99) — remain the failure mode regardless of initialisation. No pretraining strategy substitutes for examples of a class, which bounds the clinical claim that can be made from the aggregate AUROC improvements.

Robustness, including a failure of the composed arm. Table 5 reports the AUROC change under the most severe corruption. In-domain SimCLR is the most robust arm on both tasks and by a wide margin under additive noise (*-0.036* on X-ray and *-0.094* on dermatoscopy, against *-0.275* and *-0.302* for ImageNet transfer), which is consistent with its pretraining having enforced invariance to exactly such nuisance variation. The sequential ImageNet+SimCLR arm, however, collapses under severe noise on chest X-ray, falling by *0.678* AUROC to *0.309* — below chance, and the single worst behaviour we observed. An arm that is the most accurate on clean data can therefore be the least trustworthy off-distribution; reporting clean accuracy alone would have hidden this entirely.

Table 5: Change in test AUROC at the highest corruption severity relative to clean data, at the 10% label budget (mean over three seeds). More negative is worse.

Dataset	Initialisation	Gaussian noise	Brightness	Contrast	Blur
PneumoniaMNIST	from scratch	-0.098	-0.198	-0.198	-0.009
	ImageNet transfer	-0.275	-0.219	-0.116	-0.051
	in-domain SimCLR	-0.036	-0.038	-0.007	-0.007

	ImageNet + SimCLR	-0.678	-0.186	-0.001	-0.014
DermaMNIST	from scratch	-0.196	-0.327	-0.226	-0.009
	ImageNet transfer	-0.302	-0.192	-0.205	-0.087
	in-domain SimCLR	-0.094	-0.156	-0.103	-0.007
	ImageNet + SimCLR	-0.253	-0.182	-0.057	-0.059

Attribution. Figure 5 shows Grad-CAM maps [19] from the `layer3` block for melanoma test images at the 10% budget. We report a negative result honestly: at *64×64* input resolution the attributions are coarse, and for all four arms a substantial share of the evidence is drawn from peri-lesional skin rather than the lesion itself. We therefore do not claim that pretraining produces more clinically focused attention; the maps are better read as a caution that aggregate metrics at this resolution can be achieved while attending partly to context.

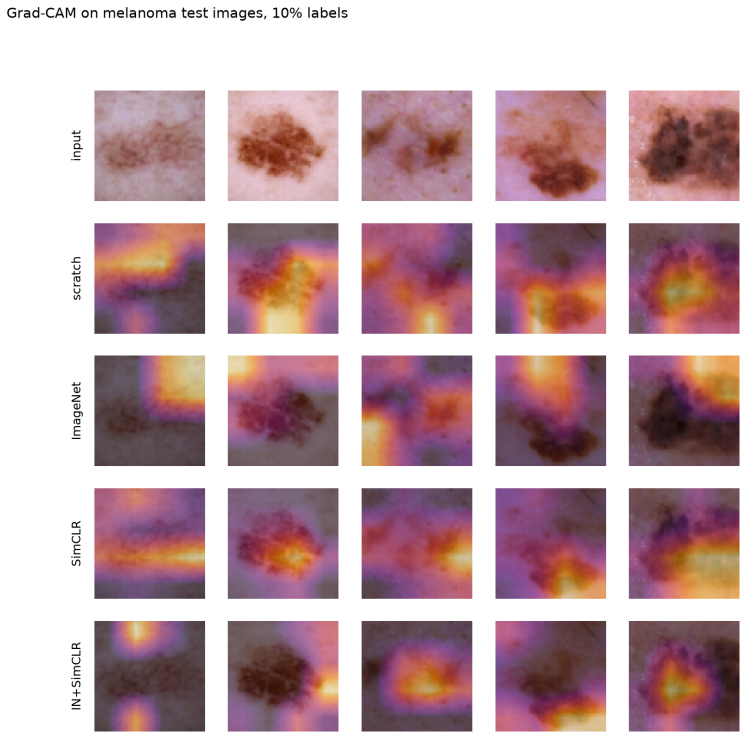


Figure 5: Grad-CAM for melanoma test images at the 10% label budget. Attributions are coarse at this input resolution and frequently emphasise tissue surrounding the lesion for every initialisation, so we draw no interpretability advantage from pretraining.

Threats to validity. Three seeds give wide intervals, so the dermatoscopy ImageNet gains (*+0.028* at 10%, *p=0.18*) are directional rather than established, and only the X-ray SimCLR effects clear *p<0.05*. Resolution is fixed at *64×64*, which limits both absolute performance and attribution quality. The unlabelled corpus is small (fewer than 7,000 images), which likely understates what in-domain self-supervision can achieve on dermatoscopy. Hyperparameters were set once per arm type from standard practice rather than tuned per cell, a deliberate trade of peak numbers for comparability. Assumption diagnostics for the paired tests were not assessed.

6 Conclusion

Holding everything but the initialisation fixed across 210 supervised runs on two modalities, we find that the choice between in-domain self-supervision and ImageNet transfer is not a matter of which method is better but of how far the target modality sits from natural images. On grayscale chest

radiography, SimCLR pretraining on fewer than 5,000 unlabelled in-domain images delivers *+0.042* AUROC over random initialisation at a 47-image label budget and yields a representation that is linearly decodable (*0.989* AUROC from a frozen backbone), while ImageNet transfer provides no significant benefit at any budget. On RGB dermatoscopy the conclusion reverses. The augmentation ablation supplies a mechanism: colour jitter, standard in SimCLR, must be removed for dermatoscopy, where colour is diagnostic, and retained for radiography, where it acts as intensity augmentation. Pretraining advantages vanish at full supervision, in-domain SimCLR is the best-calibrated and most corruption-robust arm, and no strategy rescues classes with fewer than a hundred examples.

What we take from the project is that the informative experiments were the ones designed so that a hypothesis could fail visibly in either direction, and that the results most worth reporting — the inverted colour-jitter effect and the off-distribution collapse of the composed arm — were ones no accuracy table would have surfaced. Natural extensions are a larger unlabelled corpus and higher resolution for dermatoscopy, text-supervised pretraining where reports exist [11,12], and a quantitative measure of domain distance that could predict the inversion point before any pretraining is run.

Reproducibility and code

All code, result tables and figures accompany this report. The pipeline is `run_pretrain.py` (SimCLR checkpoints), `run_sweep.py` (label-budget grid), `run_baselines.py`, `run_ablation.py`, `run_robustness.py`, `run_gradcam.py` and `make_figures.py`; shared code is in `src/`. Seeds *0,1,2* control initialisation, label subsampling and augmentation sampling. Model and training code is written against PyTorch and torchvision [5]; batched GPU augmentation uses Kornia [20]; SimCLR follows Chen et al. [6] and Grad-CAM follows Selvaraju et al. [19]; no third-party implementation was copied. Generative AI (Claude) was used for code scaffolding, debugging and draft text as described in Section 4; all experimental design decisions, validity checks and result interpretations are the authors'.

References

[1] Tschandl, Philipp, Rosendahl, Cliff, Kittler, Harald. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*. vol. 5. pp. 180161. 2018.

[2] Esteva, Andre, Kuprel, Brett, Novoa, Roberto A, Ko, Justin, Swetter, Susan M, Blau, Helen M, Thrun, Sebastian. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*. vol. 542. pp. 115-118. 2017.

[3] Rajpurkar, Pranav, Irvin, Jeremy, Zhu, Kaylie et al.. CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. *arXiv:1711.05225*. 2017.

[4] Deng, Jia, Dong, Wei, Socher, Richard, Li, Li-Jia, Li, Kai, Fei-Fei, Li. ImageNet: A large-scale hierarchical image database. *CVPR*. pp. 248-255. 2009.

[5] He, Kaiming, Zhang, Xiangyu, Ren, Shaoqing, Sun, Jian. Deep residual learning for image recognition. *CVPR*. pp. 770-778. 2016.

[6] Chen, Ting, Kornblith, Simon, Norouzi, Mohammad, Hinton, Geoffrey. A simple framework for contrastive learning of visual representations. *International Conference on Machine Learning*. pp. 1597-1607. 2020.

[7] Raghu, Maithra, Zhang, Chiyuan, Kleinberg, Jon, Bengio, Samy. Transfusion: Understanding transfer learning for medical imaging. *NeurIPS*. 2019.

[8] Esteva, Andre, Chou, Katherine, Yeung, Serena et al.. Deep learning-enabled medical computer vision. *npj Digital Medicine*. vol. 4. pp. 5. 2021.

[9] Azizi, Shekoofeh, Mustafa, Basil, Ryan, Fiona et al.. Big self-supervised models advance medical image classification. *ICCV*. pp. 3478-3488. 2021.

[10] He, Kaiming, Fan, Haoqi, Wu, Yuxin, Xie, Saining, Girshick, Ross. Momentum contrast for unsupervised visual representation learning. *CVPR*. pp. 9729-9738. 2020.

[11] Zhang, Yuhao, Jiang, Hang, Miura, Yasuhide, Manning, Christopher D, Langlotz, Curtis P. Contrastive learning of medical visual representations from paired images and text. *Machine Learning for Healthcare*. 2022.

[12] Huang, Shih-Cheng, Shen, Liyue, Lungren, Matthew P, Yeung, Serena. GLoRIA: A multimodal global-local representation learning framework for label-efficient medical image recognition. *ICCV*. pp. 3942-3951. 2021.

[13] Radford, Alec, Kim, Jong Wook, Hallacy, Chris et al.. Learning transferable visual models from natural language supervision. *ICML*. pp. 8748-8763. 2021.

[14] Kermany, Daniel S, Goldbaum, Michael, Cai, Wenjia et al.. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*. vol. 172. pp. 1122-1131. 2018.

[15] Hendrycks, Dan, Dietterich, Thomas. Benchmarking neural network robustness to common corruptions and perturbations. *ICLR*. 2019.

[16] Loshchilov, Ilya, Hutter, Frank. Decoupled weight decay regularization. *ICLR*. 2019.

[17] Yang, Jiancheng, Shi, Rui, Wei, Donglai, Liu, Zequan, Zhao, Lin, Ke, Bilian, Pfister, Hanspeter, Ni, Bingbing. MedMNIST v2 - A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*. vol. 10. pp. 41. 2023.

[18] Guo, Chuan, Pleiss, Geoff, Sun, Yu, Weinberger, Kilian Q. On calibration of modern neural networks. *ICML*. pp. 1321-1330. 2017.

[19] Selvaraju, Ramprasaath R, Cogswell, Michael, Das, Abhishek, Vedantam, Ramakrishna, Parikh, Devi, Batra, Dhruv. Grad-CAM: Visual explanations from deep networks via gradient-based localization. *ICCV*. pp. 618-626. 2017.

[20] Riba, Edgar, Mishkin, Dmytro, Ponsa, Daniel, Rublee, Ethan, Bradski, Gary. Kornia: an open source differentiable computer vision library for PyTorch. *WACV*. 2020.